

The large-field region costs more action than it has unstable directions, by the factor $2\pi^2/g^2$

The question left by the instability note has a quantitative answer, and it is favourable. The expansion inside a large-field region is **not divergent**: the configuration space is compact, so the action attains a minimum on the closed constraint set, and at that minimizer the Hessian is nonnegative on the tangent cone, so the Gaussian order is well defined. What the Nielsen–Olesen instability says is that the obvious candidate for the minimizer, a constant chromomagnetic field, is a saddle rather than a minimum. The quantitative question is then whether the unstable directions are too numerous for the Boltzmann suppression to beat, and they are not. In a constant background of magnitude $gB = \eta/\ell^2$ on a four-dimensional block of side ℓ , the count is

$$\#\{\text{unstable modes}\} = \underbrace{\frac{gB \ell^2}{2\pi}}_{\text{Landau degeneracy}} \times \underbrace{\frac{gB \ell^2}{4\pi}}_{k_3^2 + k_4^2 < gB} = \frac{(gB)^2 \ell^4}{8\pi^2} = \frac{\eta^2}{8\pi^2},$$

independent of ℓ and of the lattice spacing, exactly like the action cost $S_E/\hbar \simeq \eta^2/(4g^2)$ of the transfer note §3. Their ratio is

$$\frac{\text{action cost}}{\text{unstable-mode count}} = \frac{\eta^2/(4g^2)}{\eta^2/(8\pi^2)} = \frac{2\pi^2}{g^2},$$

uniform in η , in ℓ and in the cutoff, and large at weak coupling. So the suppression of a large-field region beats its entropy by a factor $2\pi^2/g^2$ per unstable direction, and the sum over large-field regions converges for $g^2 < 2\pi^2 \simeq 19.7$. The difficulty in the large-field region is therefore **not entropic**: it is the identification of the constrained minimizer, which the instability says is inhomogeneous. Constants explicit; the mode count is the standard one for a constant background and is a model estimate for the general case, labelled as such; nothing promoted.

1. The expansion is not divergent

Proposition 1. Let $K \subset G^\mathcal{E}$ be closed and nonempty, for instance the set where a block-averaged field strength exceeds a threshold. Then S_E attains its minimum on K , and at any minimizer U_* the second variation of S_E is nonnegative on the tangent cone of K at U_* .

Proof. $G^\mathcal{E}$ is compact and S_E continuous, so the minimum is attained; the second-order necessary condition for a constrained minimum gives nonnegativity of the Hessian on the tangent cone. $\square$

So the Gaussian integral around the constrained minimum is well defined and the expansion is not divergent at that order. The content of the Nielsen–Olesen instability is different: the constant chromomagnetic field satisfies $D^*G = 0$, hence is stationary for the unconstrained action, and has a negative Hessian direction, so it is **not** the minimizer of S_E on K . The true minimizer is inhomogeneous, and identifying it is the work.

2. Counting the unstable directions

Take a constant chromomagnetic background of magnitude B in the $(1,2)$ plane and colour direction T^3 , on a four-dimensional block of side ℓ . The charged gluon modes have frequencies

$$\omega^2 = k_3^2 + k_4^2 + (2n+1)gB \mp 2gB,$$

Landau levels in the $(1,2)$ plane and free motion in $(3,4)$, with the spin term supplied by the curvature operator of the instability note §1. The unstable modes are $n = 0$ with aligned spin and $k_3^2 + k_4^2 < gB$.

Landau degeneracy. The number of states per unit area in a level is $gB/(2\pi)$, so over the block face of area ℓ^2 it is $gB\ell^2/(2\pi)$.

Longitudinal count. The admissible (k_3, k_4) fill a disc of radius $\sqrt{gB}$, and the density of modes in a box of side ℓ is $(\ell/2\pi)^2$, giving $\pi gB\ell^2/(4\pi^2) = gB\ell^2/(4\pi)$.

Total. Multiplying, and using $gB = \eta/\ell^2$,

$$\mathcal{N}(\eta) = \frac{(gB)^2 \ell^4}{8\pi^2} = \frac{\eta^2}{8\pi^2}.$$

The count depends on η alone: a large-field region of given strength carries the same number of unstable directions at every scale, matching the scale-invariance of its action cost.

3. Cost against count

From the transfer note §3, the same region costs

$$\frac{S_E}{\hbar} \simeq \frac{1}{4g^2} \left(\frac{\eta}{\ell^2} \right)^2 \ell^4 = \frac{\eta^2}{4g^2},$$

so

Proposition 2. For a large-field region of strength η at any scale,

$$\frac{S_E/\hbar}{\mathcal{N}(\eta)} = \frac{2\pi^2}{g^2},$$

independent of η , of the block size and of the lattice spacing.

The interpretation is the usual competition between energy and entropy. Each unstable direction can be thought of as contributing at most a bounded factor to the sum over configurations in the region, so the region's weight is at most

$$e^{-\eta^2/(4g^2)} e^{C\mathcal{N}(\eta)} = \exp \left[-\frac{\eta^2}{4g^2} \left(1 - \frac{Cg^2}{2\pi^2} \right) \right],$$

negative in the exponent whenever $g^2 < 2\pi^2/C$. With C of order one this is $g^2 \lesssim 20$, which covers the whole weak-coupling regime in which the renormalization steps of the position note §6 are taken.

4. What this settles and what it leaves

Settles. The large-field region is not excluded by an entropy problem. The number of dangerous directions grows only as fast as the action cost, and the ratio is a fixed number times $1/g^2$. In particular the sum over large-field regions, with the measure bound $e^{-c\eta^2/g^2}$ of the transfer note, converges at weak coupling even after allowing a bounded factor per unstable mode. The answer to the question of the instability note §5 is therefore “merely slow”, and the slowness is quantified: a factor $g^2/(2\pi^2)$ of the exponent is spent on the unstable directions.

Leaves. The identification of the constrained minimizer. The instability says the constant field is not it, and the physical expectation is an inhomogeneous configuration, the flux-tube or “spaghetti” arrangement into which a constant chromomagnetic field decays. Expanding around it requires knowing it, or at least a lower bound on S_E over the constraint set better than the constant-field value, and that is the remaining constructive content in this corner.

A check available in principle. Proposition 2 predicts that the weight of a large-field region is $\exp[-\eta^2(1 - O(g^2))/(4g^2)]$ with the correction linear in g^2 and independent of the scale. That is a statement about the Euclidean measure that a lattice computation could test, though this repository performs none.

5. Consequence for STATE

The entropy question is answered with an explicit ratio $2\pi^2/g^2$, uniform in the strength of the region and in the scale, so the large-field sum converges at weak coupling and the expansion is slow rather than divergent. What remains in this corner is the constrained minimizer, which the Nielsen–Olesen instability shows is inhomogeneous. The next question, and the one that would close the corner, is a lower bound on the Euclidean action over the constraint set that improves on the constant-field value $\eta^2/(4g^2)$ by a factor bounded away from zero, uniformly in the scale.